

EDUCATION

Ph.D, Electrical and Computer Engineering <i>University of Windsor, ON</i>	Sept 2023 – Present
M.A.Sc, Electrical and Computer Engineering <i>University of Windsor, ON</i>	May 2019 – Dec 2021
B.Eng, Electronics and Communication Engineering <i>Anna University, India</i>	Aug 2014 – Apr 2018

EXPERIENCE

Doctoral Teaching and Research Assistant <i>University of Windsor, ON</i>	Sept 2023 – Present
Full-time Research Assistant <i>University of Windsor, ON</i>	Aug 2022 – Aug 2023
Master's Teaching and Research Assistant <i>University of Windsor, ON</i>	May 2019 – Dec 2021

RECOGNITION & AWARDS

Innovation Fellowship (CDN \$10,000) <i>University of Windsor, Canada</i> ◦ Awarded the inaugural Innovation Fellowship (top 10 of 70+ applicants) to commercialize Ph.D. research into a startup venture [Featured in UWindsor News].	May 2027
Master's Entrance Scholarship (CDN \$7500) <i>University of Windsor, Canada</i>	May 2019
Best Undergraduate Student Paper Award <i>2017 International Conference on Innovative Research in Engineering and Science, Thailand</i> ◦ Received this award for our conference paper C1	Jun 2017

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

Student Member <i>Institute of Electrical and Electronics Engineers (IEEE)</i>	Sept 2023 – Present
Professional Development Hours (PDH) <i>IEEE Educational Activities Certificates [View Certificates]</i>	2 PDH
Lithium Battery Safety <i>Lion Technology Inc.</i>	Oct 2023

SYNERGISTIC ACTIVITIES

Graduate Mentor, University of Windsor Capstone Teams <i>University of Windsor, ON</i> ◦ FSAE Accumulator Design (2025) — Ali Elshikh, Zuhair Abdo, Sahana Jayatilaka, Jared Brouerius Van Nidek ◦ SOH Indicator for a Battery Fuel Gauge (2024) — Sam Letwin, Nicolas Manzoni, Andy Garro ◦ Power System Modeling for the Formula Electric Team (2023) — Saad Arshad, Shaan Batra, Aleks Dejanovic, Luka Tojcic	Jan 2023 – Present
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- Ionic Wind Pump for Battery Cooling (2023) — Sabreen Elastal, Asra Kala, Simrat Poonia, Matthew Aguilar, Boron Al-Diraie
- Active Cell Balancing for an EV Battery Pack (2023) — Justin Haddad, Luca Mannarino, Nick Nguyen

Conference Judge, UWill Discover Student Research Conference

Mar 2026

University of Windsor, ON

Peer Reviewer

2023 – Present

15 Reviews for 5 Journals

- Applied Energy (1), Journal of Energy Storage (9), Journal of Power Sources (3), Measurement (1), MethodsX (1)

PUBLICATIONS

Journals:

- [J6] Sunil, S., Pattipati, K.R., and Balasingam, B. (Under Review). Multiple-Model Adaptive Equivalent-Circuit Identification for Li-ion Batteries via Two-Stage Least Squares. *IEEE Transactions on Transportation Electrification*.
- [J5] Sunil, S., Sundaresan, S., Pattipati, K.R., and Balasingam, B. (Accepted). The Role of Open-Circuit Voltage Characterization in Model-Based Battery Capacity Estimation. *IEEE Transactions on Instrumentation and Measurement*.
- [J4] Sunil, S., Pillai, P., Pattipati, K.R., and Balasingam, B. (2025). Two-Stage Least Squares for Equivalent-Circuit Parameter Estimation of Lithium-Ion Batteries Using Pulse-Relaxation Excitation. *IEEE Transactions on Instrumentation and Measurement*.
- [J3] Pillai, P., Desai, S., Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Least Squares Approach to Improve Standardized Estimation of OCV and Internal Resistance in Batteries. *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*.
- [J2] Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Piecewise Linear Approximation of Battery Open-Circuit Voltage Characteristics Using Dynamic Programming. *IEEE Transactions on Instrumentation and Measurement*.
- [J1] Sunil, S., Mozaffari, S., Singh, R., Shahrrava, B., and Alirezaee, S. (2023). Feature-Based Occupancy Map-Merging for Collaborative SLAM. *Sensors*, 23(6), 3114.

Conferences:

- [C11] Sunil, S., and Balasingam, B. (Accepted). An Objective Performance Evaluation of the LSTM Classifier in Time Series Classification. *2026 29th International Conference on Information Fusion*.
- [C10] Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Improved Linear Estimation of Battery Open-Circuit Voltage and RC Model Parameters from Pulse-Relaxation Data. *2025 IEEE 4th Industrial Electronics Society Annual On-Line Conference (ONCON)* (pp. 1–6).
- [C9] Nguyen, N., Sunil, S., Pillai, P., and Balasingam, B. (2025). An Offsetting-Based Correction to Improve the Accuracy of Low-Rate OCV Curves for Lithium-Ion Batteries. *2025 IEEE 34th International Symposium on Industrial Electronics (ISIE)* (pp. 1–6).
- [C8] Mahmoodzadeh, M., Sunil, S., Pillai, P., Biondi, F., Strayer, D.L., Cooper, J.M., McDonnell, A.S., and Balasingam, B. (2025). Temporal Analysis of Cognitive Workload During Manual and Partial Driving Automation Based on the Detection Response Task. *2025 IEEE 34th International Symposium on Industrial Electronics (ISIE)* (pp. 1–5).
- [C7] Sundaresan, S., Sunil, S., Balasingam, B., and Pattipati, K.R. (2023). Joint Estimation of Open-Circuit Voltage and Equivalent Circuit Model Parameters Using State-Space Model Optimization. *2023 IEEE Transportation Electrification Conference & Expo (ITEC)* (pp. 1–6).
- [C6] Sunil, S., Sundaresan, S., Pillai, P., and Balasingam, B. (2023). Inverse Characterization of Open-Circuit Voltage for State-of-Charge Estimation of Batteries. *2023 IEEE Transportation Electrification Conference & Expo (ITEC)* (pp. 1–6).
- [C5] Sunil, S., Balasingam, B., and Pattipati, K.R. (2022). State-of-Charge Estimation of Batteries Using the Extended Kalman Filter: Insights into Performance Analysis and Filter Tuning. *2022 IEEE 1st Industrial Electronics Society Annual On-Line Conference (ONCON)* (pp. 1–6).
- [C4] Sunil, S., Mozaffari, S., Singh, R., Shahrrava, B., and Alirezaee, S. (2022). LiDAR-Based Cooperative SLAM with Different Parameters. *2022 7th International Conference on Mechanical Engineering and Robotics Research (ICMERR)* (pp. 82–87).

- [C3] Sundaresan, S., Sunil, S., Balasingam, B., and Pattipati, K.R. (2022). Fast Offline Battery Capacity Estimation Approach with Performance Bounds. *2022 IEEE Electrical Power and Energy Conference (EPEC)* (pp. 188–193).
- [C2] Sunil, S., Sundaresan, S., Balasingam, B., and Pattipati, K.R. (2022). Novel Table-Based Kalman Filter for State-of-Charge Estimation of Batteries. *2022 IEEE Electrical Power and Energy Conference (EPEC)* (pp. 200–205).
- [C1] Dhaya, R., Thirumoorthy, N.R., Madhavan, S., Sunil, S., and Kanthavel, R. (2017). Resourceful and Proficient Exploitation of MDYMO in Mobile Adhoc Networks. *2017 International Conference on Innovative Research in Engineering and Science (IRES)*. (Best Student Paper Award).

Theses:

- [T2] Sunil, S. (2022). Heterogeneous Collaborative Mapping for Autonomous Mobile Systems. *University of Windsor, Canada*.
- [T1] Rishikesavan, A.S., Sunil, S., Rengaraj, S., and Kulothungan, V. (2018). LISI: Language Interpreter for Speech Impaired. *Anna University, India*.